

UNIVERSITY OF CAMBRIDGE

NATURAL SCIENCES TRIPOS Part II

Monday 8th June 2026

09:00 to 11:00

PHYSICS

PHYSICAL SCIENCES: HALF SUBJECT PHYSICS

ADVANCED AND QUANTUM OPTICS — Paper 3

*Candidates offering this paper should attempt **all six** questions (four from Section A and two from Section B).*

The approximate number of marks allocated to each question or part of a question is indicated in the right margin. This paper contains 4 sides, including this coversheet, and is accompanied by a handbook giving values of constants and containing mathematical formulae which you may quote without proof.

STATIONERY REQUIREMENTS

2 × 8-page answer booklet

Rough workpad

Yellow master coversheet

SPECIAL REQUIREMENTS

Mathematical Formulae Handbook

Approved calculator allowed

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the Invigilator.

SECTION A

Attempt **all four** questions in this Section. Answers should be concise and relevant formulae may be assumed without proof.

1 A Gaussian beam is focused by a thin lens of focal length $f = 50$ mm. The beam waist $W_0 = 2.0$ mm is located at the lens. Working at $\lambda = 800$ nm, calculate the Rayleigh range of the incident beam and show that the focused beam waist is $W'_0 \approx f\lambda/(\pi W_0)$. Hence determine W'_0 and the peak intensity at the new focus if the total beam power is $P = 5$ W.

2 Light from a thermal source with a Lorentzian spectrum of FWHM $\Delta\nu = 600$ MHz is passed into a Michelson interferometer. The fringe visibility is measured as a function of path difference Δ . Derive an expression for the visibility $\mathcal{V}(\Delta)$ and hence determine the path difference at which the visibility drops to $1/e$. If the source is replaced by a single-mode laser with linewidth 1 kHz, estimate the new $1/e$ path difference and comment on the practical implications.

3 An asymmetric planar dielectric waveguide consists of a film of refractive index n_1 and thickness d on a substrate of index n_2 , with a cover of index n_3 ($n_3 < n_2 < n_1$). Show that the condition for guiding is that the ray angle θ satisfies $\theta < \theta_c$, where $\cos \theta_c = n_2/n_1$. Explain why this waveguide, unlike the symmetric case, possesses a cut-off thickness below which no mode propagates.

4 In collinear Type-I second-harmonic generation, the two fundamental photons (frequency ω , ordinary polarisation) combine to produce a single photon at 2ω (extraordinary polarisation) in a negative uniaxial crystal ($n_e < n_o$). Write down the phase-matching condition in terms of refractive indices and show that it requires $n_o(\omega) = n_e(2\omega, \theta_{\text{pm}})$, where θ_{pm} is the angle between the propagation direction and the optic axis. Explain qualitatively why perfect phase matching is possible despite normal dispersion.

SECTION B

Attempt **both** questions from this Section.

5 Optimising the output power of a continuous-wave laser.

A four-level laser medium of length ℓ has a small-signal gain coefficient γ_0 (at the lasing frequency) and a saturation photon-flux density ϕ_s . It is placed inside a Fabry–Perot resonator of optical length d formed by one perfectly reflecting mirror ($\mathcal{R}_1 = 1$) and one output coupler of reflectance $\mathcal{R}_2 = \mathcal{R}$. Internal scattering and absorption losses are characterised by a distributed loss coefficient α_s .

- (a) Show that the total round-trip loss coefficient may be written as

$$\alpha_r = \alpha_s + \frac{1}{2d} \ln \frac{1}{\mathcal{R}}.$$

Express the threshold condition in terms of γ_0 , α_s , $\mathcal{R}$, and the lengths ℓ and d .

- (b) Using the gain-clamping condition above threshold, show that the steady-state intracavity photon-flux density is

$$\phi = \phi_s \left(\frac{\gamma_0}{\alpha_r} - 1 \right).$$

By considering the rate at which photons are lost through the output mirror, show that the output photon-flux density transmitted through the coupler is

$$\phi_{\text{out}} = \phi_s \left(\frac{\gamma_0}{\alpha_r} - 1 \right) \frac{1}{2d} \ln \frac{1}{\mathcal{R}}.$$

- (c) The output power is $P_{\text{out}} = h\nu A \phi_{\text{out}}$, where A is the beam cross-sectional area. Treating $T = 1 - \mathcal{R}$ as a continuous variable (with $T \ll 1$ so that $\ln(1/\mathcal{R}) \approx T$), show that

$$P_{\text{out}} \propto \frac{T(\gamma_0 - \alpha_s - T/2d)}{(\alpha_s + T/2d)}.$$

By differentiating and setting $dP_{\text{out}}/dT = 0$, find the optimum output-coupler transmission T_{opt} and show that it is

$$T_{\text{opt}} = 2d(\sqrt{\gamma_0 \alpha_s} - \alpha_s).$$

- (d) A Nd:YAG laser has $\gamma_0 = 0.50 \text{ m}^{-1}$, $\alpha_s = 0.02 \text{ m}^{-1}$, $d = 25 \text{ cm}$, and $\ell = d$. Calculate T_{opt} , the corresponding threshold gain, and the ratio of the optimised output power to the maximum possible intracavity power (the extraction efficiency). Comment on the physical trade-off governing the choice of $\mathcal{R}$.

6 Self-focusing and filamentation of an intense beam in a Kerr medium.

A monochromatic Gaussian beam of wavelength λ , power P , and waist radius w propagates through a medium with linear refractive index n_0 and positive Kerr coefficient n_2 (so that $n = n_0 + n_2 I$).

- (a) Write down the intensity profile $I(r)$ at the beam waist and hence show that the refractive-index profile is

$$n(r) \approx n_0 + n_2 I_0 \left(1 - \frac{2r^2}{w^2} \right)$$

near the beam axis, where I_0 is the on-axis intensity. By comparing the resulting quadratic phase with that of a thin lens, show that the Kerr medium of thickness δz acts as a lens of focal length

$$f_{\text{Kerr}} = \frac{w^2}{4 n_2 I_0 \delta z}.$$

- (b) Diffraction causes a collimated Gaussian beam to diverge with a half-angle $\theta_{\text{diff}} \approx \lambda/(\pi n_0 w)$. By requiring that the Kerr self-focusing exactly compensates the diffractive spreading, show that this balance occurs at a critical power

$$P_{\text{cr}} = \frac{\alpha \lambda^2}{4\pi n_0 n_2},$$

where α is a numerical constant of order unity that you should estimate.

- (c) For $P > P_{\text{cr}}$, the beam undergoes catastrophic self-focusing. Show that the self-focusing distance (the distance from the input at which the beam collapses) can be estimated as

$$z_{\text{sf}} \sim \frac{2n_0 w^2}{\lambda} \frac{1}{\sqrt{P/P_{\text{cr}} - 1}},$$

by considering the competition between the Kerr focal length and the Rayleigh range. For a Ti:sapphire laser pulse at $\lambda = 800$ nm with $w = 1$ mm, $P = 10$ MW, $n_0 = 1.45$, and $n_2 = 3.2 \times 10^{-20}$ m²/W, estimate P_{cr} and z_{sf} .

- (d) In practice, when $P \gg P_{\text{cr}}$ the beam does not collapse to a point. Instead, it breaks up into multiple filaments each carrying approximately P_{cr} . Explain physically why this occurs, and estimate the number of filaments produced by the beam described in part (c).

END OF PAPER